STAT6611

- Lecture 09 (T, Sept 22)
- solutions posted to Homework 1
- Homework 2 assigned due F, Oct 2 by midnight (next week)
  - practice on LSE (without constraints)
- My office hours: TW 12:30-1:30 PM & R 1-2 PM

  TA Ben: R 9:30-10:30 AM in Snedecor 2207

We now consider the least squares estimation of an estimable function $\underline{\lambda}'\underline{\beta}$ under Models IIa—VIIa.

Definition. Under Models IIa—VIIa, a least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes $Q(\underline{\beta}) \equiv (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta})$ for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

The Lagrangian function for the problem of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$, subject to the constraints $A\underline{\beta} = \underline{b}$, is

$$L(\underline{\beta}, \underline{\rho}) = Q(\underline{\beta}) + 2\underline{\rho}'(A\underline{\beta} - \underline{b}).$$

We have

$$\frac{\partial L}{\partial \underline{\beta}} = -2X'\underline{y} + 2X'X\underline{\beta} + 2A'\underline{\rho} \quad \text{and} \quad \frac{\partial L}{\partial \underline{\rho}} = 2(A\underline{\beta} - \underline{b}).$$

Equating these partial derivatives to zero gives

$$X'X\underline{\beta} + A'\underline{\rho} = X'\underline{y} \quad \text{and} \quad A\underline{\beta} = \underline{b}.$$

Thus, a necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, is that the equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix}$$

have a solution whose "first" component is $\widehat{\underline{\beta}}$. We refer to these equations as the constrained normal equations.

The following lemma can be used to establish the consistency of the constrained normal equations.

restatement of earlier result $BW=D$ iff $C(D) \subset C(B)$

Lemma 2.10. *The matrix equation $BW = D$ is consistent, that is, has a solution for W, if and only if $\mathcal{C}(D) \subset \mathcal{C}(B)$ if and only if $\mathcal{C}^{\perp}(B) \subset \mathcal{C}^{\perp}(D)$ that is, if and only if $\underline{a}'D = \underline{0}'$ for every vector $\underline{a}$ such that $\underline{a}'B = \underline{0}'$.*

($\Rightarrow$) If $D=BW$, then $\underline{a}'B=\underline{0}' \Rightarrow \underline{a}'D=\underline{a}'BW=\underline{0}'$

($\Leftarrow$) Note $(I-P_B)B=B-P_BB=B-B=0$

then $(I-P_B)D=0$ holds by assumption

So, $D=\underbrace{(I-P_B)D}_{0}+P_BD=B(B'B)^{-}B'D$

$=BW$, $W=(B'B)^{-}B'D$

Theorem 2.12. *The constrained normal equations are consistent.*

Proof. Take $\underline{a}=\begin{bmatrix}\underline{a}_1\\ \underline{a}_2\end{bmatrix}$ to be any vector such that

i.e. $\underline{a}$ is taken to be orthogonal to columns of $\begin{bmatrix}X'X & A'\\ A & 0\end{bmatrix}$

(NE) $$\underline{a}'\begin{bmatrix}X'X & A'\\ A & 0\end{bmatrix}=(\underline{a}_1',\underline{a}_2')\begin{bmatrix}X'X & A'\\ A & 0\end{bmatrix}=(\underline{a}_1'X'X+\underline{a}_2'A,\ \underline{a}_1'A')=\underline{0}'.$$

show $\underline{a}$ is also orthogonal to $\begin{pmatrix}X'\underline{y}\\ \underline{b}\end{pmatrix}$ too

By Lemma 2.10, it suffices to show that

$$\underline{a}'\begin{bmatrix}X'\underline{y}\\ \underline{b}\end{bmatrix}=(\underline{a}_1',\underline{a}_2')\begin{bmatrix}X'\underline{y}\\ \underline{b}\end{bmatrix}=\underline{a}_1'X'\underline{y}+\underline{a}_2'\underline{b}=0.$$

By (NE), $0=\underbrace{(\underline{a}_1'X'X+\underline{a}_2'A)}_{0 \text{ by (NE)}}\underline{a}_1=\underline{a}_1'X'X\underline{a}_1+\underbrace{\underline{a}_2'A\underline{a}_1}_{0 \text{ by (NE)}}=\underline{a}_1'X'X\underline{a}_1$

$\Rightarrow X\underline{a}_1=0$

$\Rightarrow \underline{a}_1'X'\underline{y}=0$ & also get by (NE) $\underline{a}_2'A=0$ too.

then $\underline{a}_2'\underline{b}=\underline{a}_2'AA^{-}\underline{b}=0$ by above. this is $\underline{b}$ since $A\underline{l}=\underline{b}$ for some $\underline{l}$

Theorem 2.13. *$Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, if and only if the constrained normal equations have a solution whose first component is $\widehat{\underline{\beta}}$.*

Proof. Let $\begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix}$ represent any solution to the constrained normal equations, and let $\widetilde{\underline{\beta}}$ represent any solution to $A\underline{\beta} = \underline{b}$.

Then, $X'X\widehat{\underline{\beta}} + A'\underline{\rho} = X'\underline{y}$, $A\widehat{\underline{\beta}} = \underline{b}$, and $A\widetilde{\underline{\beta}} = \underline{b}$. We have

$$Q(\widetilde{\underline{\beta}}) = (\underline{y} - X\widetilde{\underline{\beta}})'(\underline{y} - X\widetilde{\underline{\beta}}) = [(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'[(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]$$

$$= Q(\widehat{\underline{\beta}}) + [X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'[X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})] + 2[X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'(\underline{y} - X\widehat{\underline{\beta}}).$$

Furthermore, if $\underline{\beta}^*$ is any other value of $\underline{\beta}$ at which $Q(\underline{\beta})$ assumes a minimum, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, then $Q(\underline{\beta}^*) = Q(\widehat{\underline{\beta}})$. Thus, $[X(\widehat{\underline{\beta}} - \underline{\beta}^*)]'[X(\widehat{\underline{\beta}} - \underline{\beta}^*)] = 0$. This implies that $X(\widehat{\underline{\beta}} - \underline{\beta}^*) = \underline{0}$, or $X\underline{\beta}^* = X\widehat{\underline{\beta}}$. Then,

$$X'X\underline{\beta}^* + A'\underline{\rho} = X'X\widehat{\underline{\beta}} + A'\underline{\rho} = X'\underline{y},$$

so that $\begin{bmatrix} \underline{\beta}^* \\ \underline{\rho} \end{bmatrix}$ is also a solution to the constrained normal equations.

We now establish some properties of the coefficient matrix of the constrained normal equations.

Lemma 2.11.

$$\operatorname{rank}\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = \operatorname{rank}\begin{bmatrix} X'X \\ A \end{bmatrix} + \operatorname{rank}(A).$$

Proof. It suffices to show that $\mathcal{C}\begin{bmatrix} X'X \\ A \end{bmatrix}$ and $\mathcal{C}\begin{bmatrix} A' \\ 0 \end{bmatrix}$ are essentially disjoint, that is, have only the zero vector in common, because this implies that the union of any set of linearly independent columns of $\begin{bmatrix} X'X \\ A \end{bmatrix}$ and any set of linearly independent columns of $\begin{bmatrix} A' \\ 0 \end{bmatrix}$ is itself a set of linearly independent column vectors.

really, $\mathcal{R}(X'X)=\mathcal{R}(X)$

Lemma 2.12. *(i)* $\mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix}$ *matches* $\mathcal{R}\begin{bmatrix} X \\ A \end{bmatrix}$. *(ii)* $\operatorname{rank}\begin{bmatrix} X'X \\ A \end{bmatrix} = \operatorname{rank}\begin{bmatrix} X \\ A \end{bmatrix}$.

Note $\begin{pmatrix} X(X'X)^- & 0 \\ 0 & I \end{pmatrix}\begin{pmatrix} X'X \\ A \end{pmatrix} = \begin{pmatrix} X(X'X)^-X'X \\ A \end{pmatrix} = \begin{pmatrix} X \\ A \end{pmatrix}$ ← $P_X X$

gives $\operatorname{rank}\begin{pmatrix} X \\ A \end{pmatrix} \le \operatorname{rank}\begin{pmatrix} X'X \\ A \end{pmatrix}$

& $\begin{pmatrix} X' & 0 \\ 0 & I \end{pmatrix}\begin{pmatrix} X \\ A \end{pmatrix} = \begin{pmatrix} X'X \\ A \end{pmatrix}$

gives $\operatorname{rank}\begin{pmatrix} X'X \\ A \end{pmatrix} \le \operatorname{rank}\begin{pmatrix} X \\ A \end{pmatrix}$

Corollary to Lemma 2.11.

$$\operatorname{rank}\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = \operatorname{rank}\begin{bmatrix} X \\ A \end{bmatrix} + \operatorname{rank}(A).$$

← $\operatorname{rank}\begin{pmatrix} X'X \\ A \end{pmatrix} + \operatorname{rank}(A)$ from earlier

Theorem 2.14. *Let* $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ *represent any generalized inverse of* $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$. *Then,*

Since $\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}\begin{pmatrix} \beta \\ k \end{pmatrix} = \begin{pmatrix} X'Y \\ b \end{pmatrix}$ has sol.

then $\begin{pmatrix} \beta \\ k \end{pmatrix} = \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix}\begin{pmatrix} X'Y \\ b \end{pmatrix}$ is a sol.

(i) $A'G_{12}'A' = A'G_{21}A' = A'$;

(ii) $-A'G_{22}A = -A'G_{22}'A = X'XG_{21}'A = A'G_{21}X'X = X'XG_{12}A = A'G_{12}X'X$;

(iii) $AG_{11}A'$, $AG_{11}X'X$, *and* $X'XG_{11}A'$ *are all zero matrices;*

(iv) $X'XG_{11}X'XG_{11}X'X = X'XG_{11}X'X = X'XG_{11}'X'XG_{11}X'X = X'XG_{11}'X'X$;

(v) $\operatorname{tr}(X'XG_{11}) = \operatorname{rank}\begin{bmatrix} X'X \\ A \end{bmatrix} - \operatorname{rank}(A) = \operatorname{tr}(X'XG_{11}')$;

(vi) $X'XG_{11}X'X$ *and* $A'G_{22}A$ *are invariant for any choice of* G_{11} *and* G_{22}.

Lemma 2.13. *For any generalized inverse*

$$G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \quad \text{of} \quad \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix},$$

$X'X = X'XG_{11}X'X - A'G_{22}A.$

Theorem 2.15. *Under Models IIa—VIIa, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.*

Proof. Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ represent any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$.
By Theorem 1.3, all solutions to the constrained normal equations are

$$\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix} + \left\{ \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} - \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \right\} \underline{z}$$

$$= \begin{bmatrix} G_{11}X'\underline{y} + G_{12}\underline{b} \\ G_{21}X'\underline{y} + G_{22}\underline{b} \end{bmatrix} + \begin{bmatrix} I - G_{11}X'X + G_{12}A & -G_{11}A' \\ -G_{21}X'X + G_{22}A & I - G_{21}A' \end{bmatrix} \underline{z}$$

as $\underline{z}$ ranges over the space of all possible vectors. Thus, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions to the constrained normal equations if and only if

$$\underline{\lambda}'[I - G_{11}X'X - G_{12}A, -G_{11}A']\underline{z} = 0 \quad \text{for all } \underline{z}$$

$\Leftrightarrow \underline{\lambda}'(I - G_{11}X'X - G_{12}A) = 0 \;\&\; \underline{\lambda}'G_{11}A' = 0$

(1) $\Leftrightarrow \underline{\lambda}' = \underline{\lambda}'G_{11}X'X + \underline{\lambda}'G_{12}A \;\&\; \underline{\lambda}'G_{11}A' = 0$

(2) $\Leftrightarrow \underline{\lambda}' = \underline{q}'X'X + \underline{k}'A$ holds for some $\underline{k}$ & $\underline{q}$

$\Leftrightarrow \underline{\lambda}' \in R\begin{pmatrix} X'X \\ A \end{pmatrix} \Leftrightarrow \underline{\lambda}' \in R\begin{pmatrix} X \\ A \end{pmatrix}$

$\Leftrightarrow \underline{\lambda}'\underline{\beta}$ is estimable

Note (1) $\Rightarrow$ (2). To see (2) $\Rightarrow$ (1), let's assume (2).

Take $\underline{\lambda}'G_{11}A' = (\underbrace{\underline{g}'X'X + \underline{k}'A}_{\text{by } ②})G_{11}A'$

$= \underline{g}'\underbrace{X'XG_{11}A'}_{=0 \text{ by theorem 2.14(iii)}} + \underline{k}'\underbrace{AG_{11}A'}_{=0}$

$= 0$

$\underline{\lambda}'(G_{11}X'X + G_{12}A)$

$= (\underbrace{\underline{g}'X'X + \underline{k}'A}_{\underline{\lambda}' \text{ by } ②})(G_{11}X'X + G_{12}A)$

$= \underline{g}'X'XG_{11}X'X + \underline{g}'X'XG_{12}A +$

$+ \underline{k}'\underbrace{AG_{11}X'X}_{0 \text{ by theorem 2.14(iii)}} + \underline{k}'\underbrace{AG_{12}A'}_{A \text{ by theorem 2.14(i)}}$

$\underbrace{\qquad\qquad}_{-AG_{22}A' \text{ by theorem 2.14(ii)}}$

$= \underline{g}'(\underbrace{X'XG_{11}X'X - AG_{22}A'}_{= X'X \text{ by Lemma 2.13}}) + 0 + \underline{k}'A$

$= \underline{g}'X'X + \underline{k}'A$

$= \underline{\lambda}'$ (by ②). Hence, ② ⇒ ①